

Frontend Project Setup

A short onboarding guide for interns — Next.js + VS Code

Goal

Get a clean local frontend environment running, create a Next.js App Router project, and know what to check when VS Code or npm fails.

1. Install the essentials

- **Node.js:** use an actively supported LTS release. Next.js currently requires Node.js **20.9+**.
- **VS Code:** install the Stable release.
- **Git:** install it if the project is managed with Git.

Verify everything in a **new** terminal:

```
node -v
npm -v
git --version
```

2. Create the project

Open VS Code → **Terminal** → **New Terminal**. Move to the folder where you keep projects, then run:

```
npx create-next-app@latest my-app
cd my-app
npm run dev
```

Open **http://localhost:3000** in your browser. The current Next.js installer recommends the App Router and can configure TypeScript, Tailwind CSS, ESLint, Turbopack, and the `@/*` import alias.

Recommended intern setup

When the installer asks questions, prefer: **TypeScript = Yes, ESLint = Yes, Tailwind CSS = Yes, App Router = Yes**, and **src/ = Yes** if your team uses a src directory. Follow the existing project conventions if joining an established repository.

3. Daily workflow

Once the project is created, interns usually need only a few commands.

Start development

```
cd my-app
npm run dev
```

Then edit **app/page.tsx** (or the project's configured page) and refresh the browser. Next.js supports fast development feedback while the dev server is running.

Install an existing project

For a repository you cloned, do **not** create a new Next.js app. Instead:

```
git clone <repository-url>
cd <project-folder>
npm install
npm run dev
```

Useful checks

```
node -v
npm -v
npm run lint
npm run build
```

4. VS Code setup

- Open the **project folder**, not an individual file: **File** → **Open Folder**.
- Use the built-in terminal: **View** → **Terminal**.
- For TypeScript, use the workspace TypeScript version when the project provides one.
- Avoid installing random extensions to fix normal npm/Next.js problems; first read the terminal error.
- Keep the project-specific **package.json**, lockfile, and configuration files unchanged unless the task requires a change.

Good habit

If the team already has a **README.md**, setup script, package manager, Node version file, or **.nvmrc**, follow that project documentation first. Team conventions override this generic guide.

5. Project structure — what to recognize

Path	Purpose
app/	Next.js App Router pages, layouts, loading/error UI, and route handlers.
public/	Static files served from the site root.
src/	Optional source directory when the project is configured to use it.

Path	Purpose
package.json	Dependencies and scripts such as dev, build, start, and lint.
package-lock.json	npm dependency lockfile. Commit it when the project uses npm.
next.config.*	Next.js configuration when custom configuration is needed.
.env.local	Local environment variables. Do not commit secrets.

6. Common VS Code / npm errors

Use this checklist before asking for help. Copy the exact error message when reporting a problem.

Error / symptom	First thing to try
'node' is not recognized / command not found	Node.js is missing from PATH, or VS Code was opened before Node was installed. Install Node.js, then completely close and reopen VS Code. Verify with node -v .
'npm' is not recognized	Same PATH issue is likely. Open a fresh terminal. If Node works outside VS Code but not inside it, restart VS Code and check the terminal profile/PATH.
Cannot find module / package	From the project root, run npm install . Make sure you are in the folder containing package.json . Avoid deleting lockfiles unless the team instructs you to.
EACCES / permission denied	Do not immediately use sudo/admin commands. Check folder permissions and whether the project is inside a protected directory. On Windows, use a normal user project folder when possible.
Port 3000 is already in use	Another dev server is probably running. Stop the old process, or run Next.js on another port, e.g. npm run dev -- --port 3001 .
VS Code terminal will not open	Check the selected terminal profile and terminal.integrated settings. Try a different shell (PowerShell, Command Prompt, Bash, etc.) and restart VS Code.
TypeScript errors everywhere after opening project	Confirm VS Code opened the correct project root. Then use TypeScript: Select TypeScript Version → Use Workspace Version when appropriate.
Build works locally but differs from teammate	Compare Node.js version, package manager, lockfile, environment variables, and the exact Git commit. Run npm install only when dependency installation is needed.

When asking for help

Send: (1) exact command, (2) full error text, (3) Node/npm versions, (4) OS, and (5) what you already tried. A screenshot is useful, but text is easier to search.

7. Resource desk

Use official documentation first. These links are the preferred references for onboarding.

Next.js — Installation	https://nextjs.org/docs/app/getting-started/installation
Next.js — Documentation	https://nextjs.org/docs
Next.js — create-next-app CLI	https://nextjs.org/docs/app/api-reference/cli/create-next-app
Node.js — Downloads	https://nodejs.org/en/download
VS Code — Getting Started	https://code.visualstudio.com/docs/getstarted/overview
VS Code — Node.js tutorial	https://code.visualstudio.com/docs/nodejs/nodejs-tutorial
VS Code — Terminal troubleshooting	https://code.visualstudio.com/docs/supporting/terminal-lau nch
Git — Official site	https://git-scm.com/

8. Final onboarding checklist

- ■ Node.js installed and **node -v** works.
- ■ npm installed and **npm -v** works.
- ■ VS Code opens the correct project folder.
- ■ **npm install** completes without errors.
- ■ **npm run dev** starts the app.
- ■ **http://localhost:3000** loads.
- ■ You can edit a page and see the change.
- ■ **npm run lint** and **npm run build** are understood/working when defined by the project.
- ■ Secrets are kept out of Git.

One-line mental model

Node.js runs the tooling → **npm** installs packages → **Next.js** runs/builds the app → **VS Code** is your editor + terminal.

Prepared as a concise intern onboarding handout • Verify project-specific instructions before making changes.